

## ZANDERIJ, Suriname

WMO: 812250

Lat: 5.453N

Lon: 55.188W

Elev: 59

StdP: 14.66

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 69.5       | 70.5 | 65.2                          | 93.4 | 83.9 | 66.6 | 98.0 | 83.4 | 17.8                  | 83.0 | 16.1 | 83.4 | 0.8                   | 220  | 0.293 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 10            | 18.6                   | 93.6            | 76.3 | 93.1 | 76.4 | 91.7 | 76.6 | 79.8                | 86.0 | 79.2 | 85.7 | 78.9 | 85.5 | 7.3                  | 90   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 78.2                            | 147.0 | 82.0 | 77.4 | 142.7 | 81.0 | 77.2 | 141.9 | 80.9 | 43.3          | 86.1 | 42.8 | 86.2 | 42.3 | 86.0 | 88.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |
| 15.6              | 13.8 | 12.1 | DB                         | 66.9 | 95.9               | 1.2 | 0.9                                                | 66.0 | 96.5       | 65.3 | 97.0       | 64.6 | 97.6       | 63.7 | 98.2 |
|                   |      |      | WB                         | 65.6 | 82.2               | 2.4 | 2.0                                                | 63.9 | 83.6       | 62.5 | 84.8       | 61.1 | 85.9       | 59.4 | 87.4 |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 81.1   | 79.3  | 79.3  | 79.9  | 80.8  | 80.8  | 80.9  | 81.1  | 82.2  | 83.1  | 83.1  | 82.3  | 80.5 |
|                                                                               | DBStd       | 1.92   | 1.42  | 1.56  | 1.59  | 1.62  | 1.59  | 1.36  | 1.39  | 1.20  | 1.15  | 1.33  | 1.45  | 1.68 |
|                                                                               | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | CDD50       | 11359  | 909   | 820   | 928   | 923   | 955   | 926   | 965   | 999   | 994   | 1026  | 968   | 946  |
|                                                                               | CDD65       | 5884   | 444   | 400   | 463   | 473   | 490   | 476   | 500   | 534   | 544   | 561   | 518   | 481  |
|                                                                               | CDH74       | 49334  | 3193  | 2917  | 3676  | 3895  | 3844  | 3599  | 3950  | 4863  | 5381  | 5554  | 4732  | 3730 |
|                                                                               | CDH80       | 18693  | 979   | 898   | 1264  | 1292  | 1144  | 1147  | 1473  | 2114  | 2496  | 2584  | 2012  | 1290 |
| Wind                                                                          | WSAvg       | 4.7    | 5.1   | 5.3   | 5.4   | 4.9   | 4.4   | 3.7   | 3.5   | 4.1   | 4.6   | 5.1   | 4.8   | 4.9  |
| Precipitation                                                                 | PrecAvg     | 88.4   | 7.8   | 6.3   | 5.7   | 8.5   | 11.8  | 11.9  | 8.9   | 6.9   | 4.1   | 3.9   | 4.8   | 7.6  |
|                                                                               | PrecMax     | 115.2  | 15.9  | 21.6  | 13.1  | 24.5  | 19.1  | 19.2  | 14.2  | 11.5  | 12.6  | 11.0  | 9.6   | 18.0 |
|                                                                               | PrecMin     | 55.7   | 1.5   | 0.5   | 0.6   | 0.9   | 2.2   | 7.1   | 1.2   | 0.6   | 1.2   | 0.0   | 0.8   | 0.5  |
|                                                                               | PrecStd     | 13.6   | 3.8   | 4.0   | 3.2   | 4.8   | 3.5   | 2.6   | 3.0   | 2.7   | 2.3   | 2.6   | 1.9   | 2.8  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 89.9  | 91.0  | 91.5  | 91.8  | 91.5  | 91.4  | 91.7  | 93.5  | 94.9  | 95.2  | 94.8  | 92.1 |
|                                                                               |             | MCWB   | 76.1  | 75.3  | 76.1  | 76.5  | 77.6  | 77.1  | 76.8  | 76.8  | 76.4  | 75.8  | 75.5  | 76.5 |
|                                                                               | 2%          | DB     | 88.2  | 89.0  | 89.7  | 90.0  | 89.7  | 89.7  | 90.0  | 91.8  | 93.4  | 93.6  | 93.1  | 90.2 |
|                                                                               |             | MCWB   | 75.9  | 75.4  | 76.0  | 76.5  | 77.3  | 77.1  | 76.9  | 76.6  | 76.5  | 76.2  | 76.3  | 76.7 |
|                                                                               | 5%          | DB     | 87.3  | 87.5  | 88.0  | 89.2  | 88.0  | 88.1  | 89.5  | 91.3  | 92.8  | 93.0  | 91.5  | 89.2 |
|                                                                               |             | MCWB   | 75.8  | 75.4  | 76.1  | 76.6  | 77.6  | 77.3  | 76.9  | 76.6  | 76.5  | 76.3  | 76.7  | 76.8 |
|                                                                               | 10%         | DB     | 85.8  | 85.8  | 86.3  | 87.4  | 86.2  | 86.3  | 87.8  | 89.8  | 91.4  | 91.5  | 89.8  | 86.9 |
|                                                                               |             | MCWB   | 75.9  | 75.6  | 75.9  | 76.9  | 77.6  | 77.5  | 77.1  | 76.8  | 76.6  | 76.6  | 76.8  | 77.1 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 79.1  | 79.2  | 79.2  | 80.0  | 80.1  | 79.8  | 79.7  | 79.8  | 79.9  | 79.8  | 79.6  | 79.2 |
|                                                                               |             | MCDB   | 84.2  | 84.0  | 84.6  | 85.2  | 85.4  | 85.5  | 86.5  | 87.6  | 88.2  | 87.8  | 86.7  | 85.0 |
|                                                                               | 2%          | WB     | 78.2  | 78.2  | 78.3  | 79.2  | 79.3  | 79.0  | 78.8  | 79.0  | 79.0  | 78.9  | 78.9  | 78.7 |
|                                                                               |             | MCDB   | 83.5  | 83.3  | 84.1  | 84.6  | 84.7  | 84.8  | 85.5  | 87.0  | 87.5  | 87.2  | 86.3  | 84.6 |
|                                                                               | 5%          | WB     | 77.4  | 77.5  | 77.7  | 78.5  | 78.8  | 78.4  | 78.1  | 78.3  | 78.3  | 78.3  | 78.3  | 78.2 |
|                                                                               |             | MCDB   | 83.0  | 82.9  | 83.9  | 84.0  | 84.2  | 84.2  | 84.9  | 86.4  | 86.8  | 86.6  | 85.7  | 84.2 |
|                                                                               | 10%         | WB     | 76.8  | 76.8  | 77.0  | 77.9  | 78.2  | 78.0  | 77.6  | 77.8  | 77.8  | 77.8  | 77.9  | 77.6 |
|                                                                               |             | MCDB   | 82.5  | 82.3  | 83.5  | 83.4  | 83.4  | 83.6  | 84.5  | 85.8  | 86.3  | 86.1  | 85.3  | 83.8 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 14.3  | 14.1  | 15.1  | 14.3  | 13.3  | 14.2  | 15.8  | 17.3  | 18.6  | 18.6  | 17.3  | 14.9 |
|                                                                               |             | MCDBR  | 16.7  | 17.3  | 17.2  | 17.1  | 15.3  | 15.7  | 17.1  | 18.4  | 19.4  | 19.7  | 18.8  | 17.3 |
|                                                                               |             | MCWBR  | 6.2   | 6.2   | 6.0   | 5.7   | 5.4   | 5.6   | 5.7   | 5.5   | 5.3   | 5.1   | 5.4   | 5.6  |
|                                                                               | 5% WB       | MCDBR  | 13.6  | 13.1  | 14.3  | 13.6  | 13.3  | 14.3  | 15.8  | 17.1  | 18.3  | 18.4  | 17.1  | 14.6 |
|                                                                               |             | MCWBR  | 5.9   | 5.8   | 5.7   | 5.4   | 5.3   | 5.7   | 6.0   | 5.9   | 5.7   | 5.6   | 5.7   | 5.8  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.425  | 0.457 | 0.458 | 0.477 | 0.452 | 0.399 | 0.399 | 0.400 | 0.412 | 0.405 | 0.394 | 0.446 |      |
|                                                                               | taud        | 2.426  | 2.314 | 2.317 | 2.256 | 2.355 | 2.527 | 2.515 | 2.502 | 2.450 | 2.485 | 2.516 | 2.362 |      |
|                                                                               | Ebn at Noon | 284    | 277   | 276   | 266   | 266   | 278   | 279   | 284   | 285   | 289   | 291   | 275   |      |
|                                                                               | Edh at Noon | 37     | 43    | 43    | 45    | 40    | 33    | 33    | 35    | 37    | 36    | 34    | 39    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 1409   | 1429  | 1535  | 1503  | 1437  | 1541  | 1711  | 1901  | 1990  | 1905  | 1670  | 1406  |      |
|                                                                               | RadStd      | 97     | 91    | 66    | 106   | 84    | 77    | 57    | 70    | 48    | 71    | 94    | 105   |      |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page